Overview of DAWN Use Cases

Daniel King (Old Dog Consulting)

Kehan Yao (China Mobile)

Ken Adler (Indeed)

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https://datatracker.ietf.org/doc/draft-kay-dawn-use-cases/

Problem Statement

- Fragmented ecosystems & cross-domain invisibility
- Static configurations don’t scale for large clusters
- Diverse entities & dynamic attributes

Next Step: Let’s explore concrete use cases to define the essential capabilities a modern and common discovery mechanism must provide.

Four Categories of DAWN Use Cases

01. Capability-Oriented

Find entities by function/capability
(who can do this)

02. Resource-Oriented

Find entities by resource attributes
(where are suitable resources)

03. Administrative Scope Extensions

- Local (link/segment/physical space) discovery
- Cross-organization; cross-tenant discovery

04. Operational Discovery

Operations, audit, troubleshooting, compliance,etc.

- The first two categories are foundational discovery use cases, while the latter two represent extensions or deployment variants.

1. Capability-oriented Discovery

Core Question: What specific actions, functions, or capabilities can an entity perform?

Description: Alice, an employee at Company A, needs to create a year-end summary slide deck.

Discovery Query:

"Help me design a 10-slide summary with charts and illustrations."

"Entity Type: Agent/ Tool"
"Capability: Text-to-Image"
"Supported Protocol: HTTPS"
...

Alice → **Alice's Personal AI Assistant** → **Chart-Gen Agent/Tool**

(User) (Discovering Entity) (Discovered Entity)

- Multi-attribute Search
- Return sufficient metadata *(Endpoint URI, capability card, input/output format, etc)*

▶ While "capability" defines what an entity can do, discovery often needs to go further. In some cases, Service deployers/providers want to discover resources that can implement these functions/capabilities.

2. Resource-oriented Discovery

Core Question: What can be used to implement these functions, and where are they?

Description: Alice asks Cloud Orchestrator Agent to help deploy specific models and tasks.

Discovery Query:

"I need to deploy two model variants (60B and 1000B parameters) and run 100 rendering tasks across them."

"GPU_Memory > 200GB"
"GPU_Type: A"
"Current_Load: <70%"
...

- ➢ Static Properties *(hardware specs, etc..)*
- ➢ Dynamic attributes *(Current_Load, Availability, etc..)*

Alice

(User)

Cloud Orchestrator Agent

(Discovering Entity)

Compute Clusters

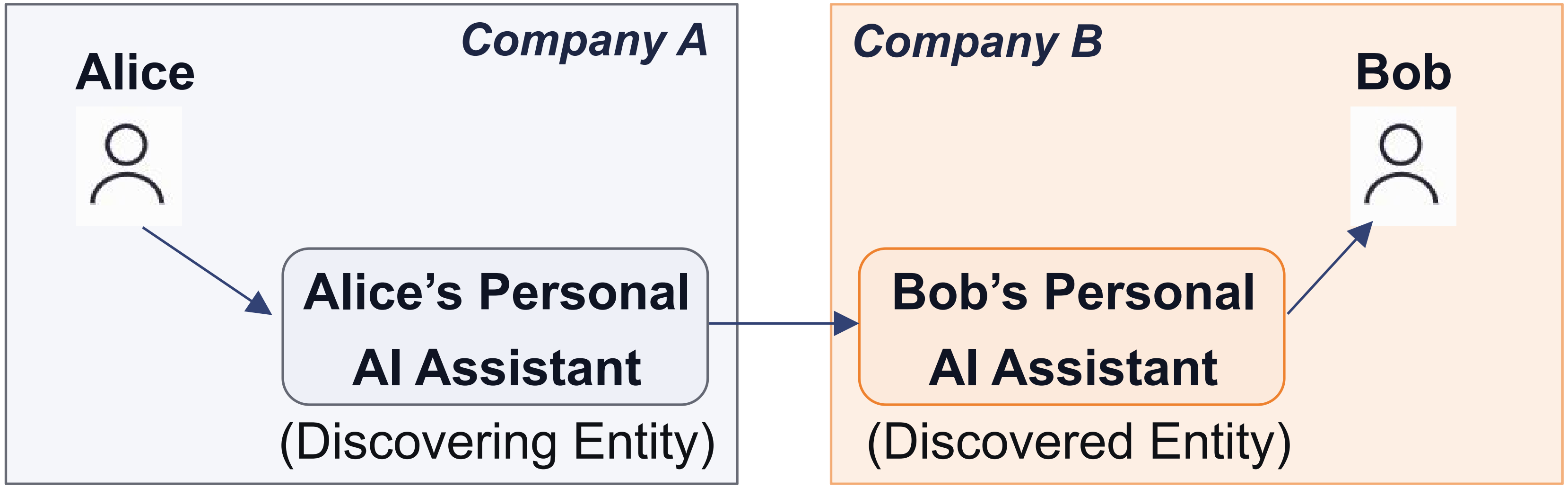
(Discovered Entity)

▶ The aforementioned “capability” search and “resource” search happen within a single administrative domain (Company A), but can extends to cross organizational boundaries.

3. Administrative Scope Extensions

Core Question: How does discovery work across organizations/tenants?

Description: Alice needs to schedule an online meeting with Bob (client customer) from Company B.



Discovery Query: "Target_domain: Company B" or
"Target_endpoint: Bob's assistant's URI"

- ➢ Trust indicators and delegation

(Need authentication of discovery information.

Authentication of **discovered entity is out-of-scope**.)

- ➢ Decentralized architectures

▶ Discovery can also happen in OPS scenarios, which share similar patterns and characteristics.

4. Operational Discovery

Core Question: How does discovery support operations and troubleshooting?

Description: Alice, acting as a system administrator, asks the OAM agent for fault localization.

"Show me fault localization, system logs, and root cause analysis for the last 2 hours."

Discovery Query:

"Telemetry_Source": Source = "Network_Edge"
"Log_Indexer": Timeframe = "Last_2h"
"RCA_Engine": Capability = "Anomaly_Detection"

Alice

(Operator OR Discovering Entity)

OAM Agent

(Discovering Entity)

- **Telemetry Source**
- **Log Indexer**
- **RCA Engine**
- **etc..**

(Discovered Entity)

- ➢ Human can also initiate discovery
- ➢ Discovery objective for audit trail
- ➢ Share similar discovery requirements

From Use Cases to Requirements Categorization

Use Case Observation	Core Requirement Category
Diverse entity types (Agents, Tools, Workloads, Services, Functions)	**Entity Classification** (REQ-CLASS-*)
Discovery of capabilities, resources, and cross-domain context	**Discovery Actors & Scenarios** (REQ-DISC-*)
Hybrid of static metadata and dynamic operational properties	**Entity Properties Model** (REQ-PROP-*)
Cross-organizational collaboration and decentralized networks	**Scalability & Distributed Architecture** (REQ-ARCH-*)
Need for trustworthy, verifiable, and secure discovery data	**Trust & Security** (REQ-SEC-*)